

Problem 109. Let a, b, c, d be positive real numbers such that $2(a + b + c + d) \geq abcd$.

Prove that

$$a^2 + b^2 + c^2 + d^2 \geq abcd.$$

Solution.

Suppose on the contrary that $abcd > a^2 + b^2 + c^2 + d^2$. By the QM-AM inequality, we have

$$2(a + b + c + d) \geq abcd > a^2 + b^2 + c^2 + d^2 \geq \frac{(a + b + c + d)^2}{4},$$

which gives $a + b + c + d < 8$. Also, by the AM-GM inequality, we have

$$abcd > a^2 + b^2 + c^2 + d^2 \geq 4\sqrt[4]{a^2b^2c^2d^2} = 4\sqrt{abcd},$$

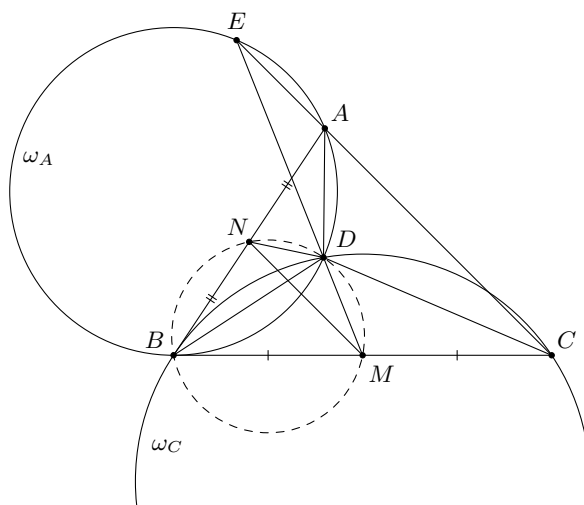
which gives $abcd > 16$. This yields the contradiction

$$16 > 2(a + b + c + d) \geq abcd > 16.$$

Therefore, we must have $a^2 + b^2 + c^2 + d^2 \geq abcd$. □

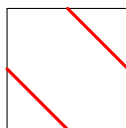
Problem 110. In $\triangle ABC$, the circle ω_A through A is tangent to line BC at B . The circle ω_C through C is tangent to line AB at B . Let ω_A and ω_C meet again at D . Let M be the midpoint of line segment BC , and let E be the intersection of lines MD and AC . Show that E lies on ω_A .

Solution.



Let N be the midpoint of AB . By tangential angles, $\angle CBD = \angle BAD$ and $\angle DBA = \angle DCB$, so triangles DAB and DBC are similar. By definition of the midpoints M, N , so are triangles AND and BMD . In particular, $\angle DNA = \angle DMB$, so $BNDM$ is cyclic. But MN and AC are parallel by the midpoint theorem, so $\angle DBA = \angle DBN = \angle DMN = \angle EMN = \angle MEA = \angle DEA$, hence $EADB$ is cyclic, completing the proof. □

Problem 111. Let N be a positive integer. A collection of $4N^2$ unit tiles with two red segments: one from the midpoint of the bottom side to the midpoint of the left side, and the other from the midpoint of the right side to the midpoint of the top side. A unit tile is shown below.

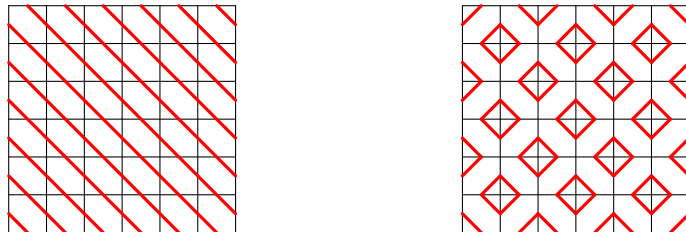


These tiles are assembled into a $2N \times 2N$ board. Moreover, these tiles can be rotated. The red segments on the tiles then define paths on the board.

Determine the least possible number and the largest possible number of such paths.

Solution. Let M denote the number of paths. Notice that there are two types of paths: (1) those that start and end at a point on the boundary of the board and (2) closed paths in the interior of the board. Let m_1, m_2 denote the respective numbers of paths of either type. There are $8N$ points on the boundary of the board, and each of these is the starting point or endpoint of exactly one path, so $m_1 = 4N$. Trivially, $m_2 \geq 0$, so $M = m_1 + m_2 \geq 4N$.

The paths on the board are made up of $8N^2$ segments in total. There are only 4 possible paths of one segment, in the corners of the board. All other paths on the boundary of the board therefore consist of at least 2 segments. Moreover, all closed paths in the interior of the board consist of at least 4 segments. Hence $8N^2 \geq 4 \cdot 1 + (m_1 - 4) \cdot 2 + m_2 \cdot 4$, which gives $m_2 \leq N^2 + (N - 1)^2$, so $M = m_1 + m_2 \leq N^2 + (N + 1)^2$. We have thus shown that $4N \leq M \leq N^2 + (N + 1)^2$. These minimum and maximum values can indeed be attained, as shown below for $N = 3$.



The constructions generalize easily. The constructions clearly attain the required bounds because they satisfy the equalities in our arguments showing the two bounds. Indeed the board on the left has $m_2 = 0$ so give the lower bound. The one on the right has exactly 4 boundary paths with one segment, all other $4N - 4$ boundary paths with 2 segments, and all remaining paths with 4 segments. $\square$

Problem 112. A divisor d of a positive integer n is said to be a *close divisor* of n if $\sqrt{n} < d < 2\sqrt{n}$.

Does there exist a positive integer with exactly 2020 close divisors?

Solution 1. Let m be an odd integer with exactly 2020 positive divisors and which is (automatically) not a square. For example, $m = 3^{2019}$ suffices, but of course there are many alternatives. Now consider $n = 2^k m$, for some integer k such that $2^k > m$. Any divisor of n is then of the form $2^\ell d$ where d is a divisor of m . We will now show that for every such divisor d , there exists a unique ℓ such that $2^\ell d$ is a close divisor. Because $\sqrt{n}$ is not an integer, there certainly is a unique integer a such that $\sqrt{n} < 2^a d < 2\sqrt{n}$. Because $2^k > m$, we have $m < \sqrt{n} < 2^k$. Combining with $1 \leq d \leq m$, we find

$$1 < \frac{\sqrt{n}}{d} < 2^a < \frac{2\sqrt{n}}{d} < 2 \cdot 2^k$$

and so we see that $0 < a \leq k$. So $2^a d$ is indeed a close divisor. Because m has exactly 2020 divisors, we find that n has exactly 2020 close divisors. $\square$

Solution 2. We prove by induction on k that there exists a perfect square with exactly k close divisors. For the base case $k = 1$, the number $6^2 = 36$ has exactly one close divisor 9.

Assume m^2 is a perfect square with exactly k close divisors. By Bertrand's postulate, there exists a prime p satisfying $m < p < 2m$. We claim that $n = (pm)^2$ has exactly $k + 1$ close divisors. Consider any divisor d . Note that d is a close divisor if and only if $pm < d < 2pm$.

- If d is not divisible by p , then $d \leq m^2 < pm$. So d is not a close divisor.
- If $d = pb$ where $(b, p) = 1$, then b is a divisor of m^2 . As $pm < pb < 2pm$ if and only if $m < b < 2m$, d is a close divisor of n if and only if b is a close divisor of m^2 . As there are exactly k close divisors of m^2 , there are exactly k such close divisors d of n .
- If $d = p^2$, the relation $m < p < 2m$ implies $pm < p^2 < 2pm$, so d is a close divisor.
- If $d = p^2 c$ where $c \geq 2$ and c divides m^2 , then $d \geq 2p^2 > 2pm$. So d is not a close divisor.

Therefore, n has exactly $k + 1$ close divisors. The claim holds by induction. $\square$